Beyond Heuristic Repair: When Can Exact Optimization Recover Table Structure?

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Abstract. Modern Table Structure Recognition (TSR) systems often use post-processing heuristics to convert local predictions into structurally valid tables. However, these procedures can alter or discard predictions produced by the underlying model. We study this behavior across four TSR paradigms—Split-and-Merge, Sequence Generation, Object Detection, and Relation Learning—using two hypotheses: whether heuristic reconstruction destroys locally plausible predictions (H1), and whether exact combinatorial optimization can recover such predictions (H2). Across the four paradigms, we identify distinct mechanisms by which valid predictions are lost during reconstruction or detection. Replacing the relevant repair stages with exact set-packing optimization also produces different outcomes. On SPLERGE, the optimization recovers 632 of 634 destroyed true positives (99.68%). On TATR, a confidence-based objective cannot improve on greedy NMS because 99.8% of its conflict components are simple pairwise conflicts; an alternative geometric objective produces only a net gain of one true positive. On TableFormer, correcting a conflict-definition artifact reduces an initial apparent performance loss to a statistically indistinguishable result from the raw predictions ($p = 0.320$). For LORE-TSR, the proposed optimization is not applicable because the relevant failures occur before a reconstruction stage. These results indicate that exact optimization is useful when the prediction space provides a discrete set of candidates and an evaluation-independent, content-aware signal is available for selecting among them. The study therefore separates cases where optimization can recover structure from those where the underlying prediction representation or objective limits what post-processing can achieve.

Keywords: Table Structure Recognition · Post-Processing Heuristics · Combinatorial Optimization · Set Packing · Document Understanding

1 Introduction

Document understanding pipelines rely on Table Structure Recognition (TSR) to convert document images into structured representations of rows, columns, cells, and their relationships [10,17]. TSR systems use different prediction paradigms, including grid-based split-and-merge methods, sequence generation, object detection, and relation or logical-coordinate prediction. Despite these architectural

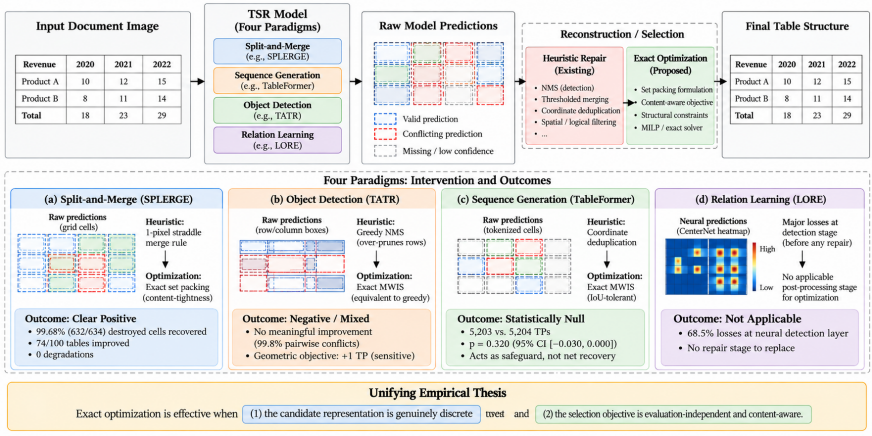


Fig. 1: Overview of the proposed analysis across four TSR paradigms. We trace how raw predictions are transformed by reconstruction or filtering stages (H1) and investigate whether exact set-packing optimization can replace applicable heuristic selection stages (H2). The four paradigms produce different outcomes, revealing conditions that determine when optimization can recover structure.

differences, the final predictions often require a reconstruction or filtering stage to obtain a structurally consistent table.

These post-processing stages are usually implemented with deterministic rules. Depending on the prediction paradigm, they may include non-maximum suppression (NMS) for overlapping detections [10], threshold-based merging of adjacent regions [1,15], coordinate deduplication in sequence-generation pipelines [6], or confidence and logical-coordinate filtering in relation-based systems [13, 14]. Such procedures are intended to remove conflicts and produce valid table structures. However, a rule that resolves a structural conflict does not necessarily preserve the correct prediction. A locally plausible prediction may satisfy the model's output semantics while being removed by a subsequent heuristic because of overlap, confidence, or geometric thresholds.

This raises a basic question about TSR pipelines: *when a post-processing stage changes the model's predictions, does it merely repair invalid structure, or can it also destroy valid structure that was already present in the raw predictions?* We study this question by tracing the transition from raw model outputs to reconstructed table structures rather than evaluating only the final end-to-end prediction. We further ask whether exact combinatorial optimization can replace such heuristic selection when the set of candidate predictions and an appropriate objective are available.

We conduct a systematic, source-traced empirical analysis across four TSR paradigms: Split-and-Merge (SPLERGE [1, 15]), Sequence Generation (TableFormer [6]), Object Detection / DETR (TATR [10]), and Relation Learning / Logical Coordinate Regression (LORE [13]). We formulate two hypotheses:

- **Hypothesis 1 (H1 — Heuristic Destruction):** Raw TSR predictions can contain locally plausible structural predictions that are subsequently discarded or altered by heuristic reconstruction or filtering stages.
- **Hypothesis 2 (H2 — Exact Optimization Recovery):** For applicable reconstruction stages, replacing heuristic selection with an exact, content-aware combinatorial optimization layer can recover valid structure while satisfying the same structural consistency constraints.

The four paradigms produce substantially different outcomes. In SPLERGE, an unconditional 1-pixel straddle rule removes a substantial fraction of true-positive cells; replacing the merge heuristic with exact set packing and a content-tightness objective recovers 632 of 634 destroyed true positives (99.68%), with 74 of 100 evaluated tables improving and none degrading. In TATR, row-level NMS removes valid adjacent rows. Under a model-confidence objective, exact optimization provides no meaningful improvement because 99.8% of conflict-graph connected components are simple pairwise conflicts, for which greedy NMS is already optimal. A geometric objective produces a net gain of only one true positive and introduces new errors associated with sub-pixel coordinate variation. In TableFormer, an initial apparent loss of 190 true positives is traced to the definition of conflicts rather than to the optimization itself. After introducing IoU-tolerant conflict criteria, exact optimization produces a result statistically indistinguishable from the raw predictions (5,203 vs. 5,204 true positives; $p = 0.320$; bootstrap 95% CI $[-0.030, 0.000]$). Finally, LORE represents a different failure mode: the principal losses occur at the neural detection stage, before a reconstruction stage exists, making the proposed optimization intervention inapplicable.

Taken together, these results suggest that the usefulness of exact optimization depends on properties of both the prediction space and the selection objective. The positive SPLERGE result is associated with a discrete candidate representation together with an external, content-aware signal, whereas TableFormer's continuous coordinate representation limits what a selection-based optimizer can recover. TATR further shows that an optimization procedure cannot improve on a greedy method when the conflict structure and objective make the greedy solution already optimal. These observations support a unifying empirical thesis: exact optimization is most useful when the candidate representation is genuinely discrete and the selection objective provides information about correctness that is independent of the model's own potentially miscalibrated confidence scores.

1.1 Summary of Main Contributions

- **Four-Paradigm Analysis:** We trace raw predictions through reconstruction or filtering across four TSR paradigms and identify mechanisms that remove valid predictions.
- **Exact Optimization:** We formulate applicable reconstruction stages as exact set-packing problems using objectives derived from model or document information.

- **Recovery Conditions:** Our experiments identify a discrete candidate representation and an evaluation-independent, content-aware signal as the two properties associated with successful recovery.
- **Measurement Audit:** We identify and correct methodological artifacts in the TableFormer and SPLERGE experiments.

2 Related Work

Table Structure Recognition (TSR) converts document images containing tables into machine-readable structural representations, such as HTML or logical cell coordinates [10, 17]. Existing methods use different representations of table structure. In this work, we organize representative approaches into four families: split-and-merge, sequence generation, object detection, and logical/relation-based prediction. We also review the post-processing operations used by these methods, since these operations form the focus of our analysis.

2.1 Table Structure Recognition Paradigms

Split-and-Merge. Split-and-merge methods first divide a table into a basic grid and then merge grid elements to recover cells that span multiple rows or columns. SPLERGE [11] uses separate split and merge networks, with projection pooling for grid prediction and grid pooling for merging. Later methods such as SEM [15], TSRFormer [4], TRUST [2], GridFormer [5], and related approaches improve the detection of separation lines, grid elements, or merge decisions. This family provides an explicit discrete representation of the table grid, but the final structure still depends on the rules used to convert predicted grid elements into cells.

Sequence Generation. Image-to-sequence methods formulate TSR as the generation of a structured markup sequence. PubTabNet and the Encoder-Dual-Decoder framework established a large-scale benchmark and an image-to-markup formulation for TSR [17]. TableFormer [6] replaces recurrent decoding with Transformer-based decoding and associates generated structural elements with image regions. Other approaches, including LGPMA [7] and VAST [3], introduce additional mechanisms for modeling local structure and visual alignment. Because sequence-based predictions may contain duplicate or spatially inconsistent elements, practical pipelines may apply coordinate filtering, deduplication, or alignment with OCR-derived information after decoding.

Object Detection. Detection-based approaches represent table elements as objects and predict their spatial coordinates. Early methods used conventional object detectors for rows, columns, or cells, including DeepDeSRT [9], TabStructNet [8], and GTE [16]. PubTables-1M introduced a large-scale benchmark together with the Table Transformer (TATR), a DETR-based model that predicts table rows, columns, and cells directly from learned queries [10]. More recent work, such as TableCenterNet, jointly predicts spatial and logical cell coordinates

in a one-stage architecture [12]. Since these approaches operate on continuous bounding boxes or point locations, duplicate and overlapping predictions are commonly handled using confidence thresholds or non-maximum suppression.

Relation and Logical-Coordinate Prediction. Another family represents table structure through relationships between detected cells or through logical row and column coordinates. GraphTSR predicts relations between detected cells and introduced the SciTSR benchmark for complex tables [1]. TGRNet similarly models relationships between table cells using graph-based reasoning [14]. LORE instead predicts both the spatial location and logical coordinates of cells using an anchor-free detection framework [13]. These approaches differ from split-and-merge methods in that the logical structure is predicted more directly rather than reconstructed solely from a predicted grid.

2.2 Post-Processing in TSR

Although TSR architectures differ substantially, many pipelines contain a stage that converts raw predictions into the final table structure. The operation depends on the prediction representation. Detection-based methods may use non-maximum suppression or confidence filtering, sequence-based methods may remove duplicate or misaligned predictions, and split-and-merge methods may merge neighboring grid elements according to geometric rules. Such operations are useful for enforcing structural consistency, but they also modify the predictions produced by the underlying model.

This distinction is important for evaluating TSR systems. Standard metrics such as Tree-Edit-Distance-based Similarity (TEDS) [17] and GriTS [10] evaluate the final reconstructed structure. Consequently, a correct prediction that is removed during post-processing is indistinguishable from a prediction that was never produced by the model when only the final output is considered. Our study therefore traces predictions before and after reconstruction to measure this difference directly.

2.3 Combinatorial Optimization for Structural Selection

Combinatorial optimization provides an alternative to local selection rules when multiple candidate predictions are mutually incompatible. In particular, set-packing and maximum-weight independent-set formulations can select a globally consistent subset of candidates subject to pairwise or higher-order constraints. In our setting, each candidate structural element is associated with a binary decision variable, while the optimization objective determines which candidates are retained.

The important distinction from heuristic post-processing is that an exact solver considers the candidate set jointly rather than making independent local decisions. However, exact optimization does not by itself guarantee improved recognition accuracy. Its effectiveness depends on both the candidate representation and the quality of the objective used to rank competing candidates. This

observation motivates our evaluation of exact optimization within multiple TSR prediction paradigms.

2.4 Positioning of Our Study

Our study focuses on the transition between raw TSR predictions and final reconstructed structures. Across four TSR paradigms, we trace predictions that are removed during post-processing and, where an explicit candidate set exists, replace the corresponding heuristic selection with exact optimization. This enables us to separate errors introduced by reconstruction from errors already present in the model output and to examine how candidate representation and objective design affect optimization-based recovery.

3 Methodology

We investigate two questions across four Table Structure Recognition (TSR) paradigms. First, do post-processing stages remove structurally correct predictions produced by the underlying model? Second, when such predictions are available, can an exact optimization procedure retain them while satisfying the structural constraints imposed by the pipeline? We refer to these questions as H1 and H2, respectively.

Our methodology separates the raw predictions produced by each model from the subsequent reconstruction or filtering stage. This separation allows us to trace where predictions are lost and, where applicable, replace the original heuristic with an exact optimization problem defined over the same candidate set.

3.1 Research Questions and Hypotheses

We evaluate the following hypotheses:

- **Hypothesis 1 (H1 – Heuristic Destruction):** Post-processing stages can discard candidate predictions that are correct with respect to the ground-truth table, rather than only removing incorrect predictions.
- **Hypothesis 2 (H2 – Exact Optimization Recovery):** When correct candidates remain available in the raw prediction set, replacing a heuristic selection stage with an exact combinatorial optimization procedure can recover some of the predictions removed by the original post-processing stage.

H2 is evaluated only for paradigms in which the relevant post-processing stage operates on an explicit candidate set. LORE-TSR is therefore treated as a boundary case. In LORE, a substantial fraction of cell failures occurs during the CenterNet detection stage, before a candidate is passed to a subsequent structural reconstruction stage. Such failures cannot be recovered by a post-processing optimizer because the corresponding candidate is absent from the candidate set.

3.2 Common Evaluation Framework

Although the four TSR systems use different prediction representations, we apply the same analysis at the pipeline level:

$$C_{\text{raw}} \xrightarrow{P} C_{\text{final}}. \tag{1}$$

Here, C_{raw} denotes the predictions emitted by the model before post-processing, and P denotes the original reconstruction or filtering operation.

For each prediction, we record whether it is retained, removed, or modified by P. Predictions are then matched against the corresponding ground-truth structure using the evaluation protocol of the underlying TSR system. This allows us to distinguish errors introduced during post-processing from errors already present in the raw model output.

For each experiment, we measure the following quantities where applicable:

- TP_{raw}: true positives in the raw model predictions;
- TP_{final}: true positives remaining after the original post-processing stage;
- $\text{TP}_{\text{destroyed}}$: true positives present in the raw predictions but removed by post-processing;
- TP_{opt}: true positives obtained after replacing the relevant heuristic with exact optimization.

The exact matching criterion is kept consistent with the evaluation protocol of each experiment. We do not modify ground-truth information or use ground-truth annotations to construct optimization objectives.

3.3 Tracing Prediction Loss

For a given model, let

$$C_{\text{raw}} = \{c_1, c_2, \ldots, c_N\} \tag{2}$$

denote the candidates produced before post-processing. Depending on the TSR paradigm, a candidate may represent a grid cell, a bounding box, or a decoded structural element.

The original post-processing stage produces

$$C_{\text{final}} = P(C_{\text{raw}}). \tag{3}$$

A raw candidate is considered *destroyed* when it satisfies the evaluation criterion for a true positive in the raw prediction set but is absent from the final prediction set. Thus, the number of destroyed true positives is measured by comparing the matched true-positive predictions before and after post-processing:

$$\text{TP}_{\text{destroyed}} = \text{TP}_{\text{raw}} - \text{TP}_{\text{retained}}. \tag{4}$$

This definition is applied using the appropriate matching procedure for each paradigm rather than imposing a single geometric criterion on all prediction

types. For example, bounding-box overlap is used for detection-style predictions, whereas grid-cell or structural matching is used where the prediction representation is discrete.

This source-level tracing is important because a final false negative can arise in two different ways: the model may never have produced the corresponding structure, or the model may have produced it and the subsequent processing stage may have removed it. Our analysis distinguishes these cases.

3.4 Exact Optimization Formulation

For paradigms in which post-processing selects a subset of mutually incompatible candidates, we replace the original heuristic with a binary set-packing formulation.

Let $x_i \in \{0, 1\}$ indicate whether candidate c_i is retained. Each candidate is assigned a non-oracle weight w_i that can be computed from information available to the model or the document representation, such as model confidence, geometric regularity, or OCR-based content coverage.

We solve

$$\max_{\mathbf{x}} \sum_{i=1}^{N} w_i x_i \tag{5}$$

subject to

$$x_i + x_j \leq 1, \qquad \forall (i, j) \in \mathcal{E}, \tag{6}$$

and

$$x_i \in \{0, 1\}. \tag{7}$$

Here, $\mathcal{E}$ is the set of pairs of candidates that cannot simultaneously be selected under the structural constraints of the corresponding TSR system.

Equivalently, the problem can be viewed as a Maximum Weight Independent Set (MWIS) problem over a conflict graph whose vertices are the raw candidates and whose edges represent incompatible predictions.

All optimization problems are solved to global optimality using the HiGHS mixed-integer programming solver through `scipy.optimize.milp`.

An important limitation follows directly from this formulation:

$$C_{\text{opt}} \subseteq C_{\text{raw}}. \tag{8}$$

Therefore, optimization can change which existing candidates are retained, but it cannot recover a structure that was never represented in the raw candidate set. This distinction is central to interpreting the results for the four TSR paradigms.

Table 1: Four-paradigm intervention matrix. The table summarizes the prediction representation, original post-processing operation, and corresponding optimization intervention.

Paradigm	Model	Raw representation	Original processing	Optimization
Split-and-Merge	SPLERGE	Discrete grid cells	1-pixel straddle merge	Set packing
Object Detection	TATR	Bounding boxes	Greedy NMS	MWIS
Sequence Generation	TableFormer	Continuous cell boxes	Coordinate deduplication	IoU-tolerant MWIS
Relation Learning	LORE-TSR	CenterNet predictions	Detection threshold	Not applicable

3.5 Paradigm-Specific Interventions

The candidate representation, original post-processing operation, and optimization objective differ across the four paradigms. Table 1 summarizes the interventions.

Split-and-Merge (SPLERGE) SPLERGE represents a table using a planar grid and subsequently merges neighboring grid elements to form spanning cells. In our implementation, the original post-processing stage merges adjacent elements when the predicted separation boundary falls within a 1-pixel tolerance of the cell boundary.

We replace this rule with an exact set-packing formulation. Candidate cells are assigned weights based on OCR token coverage, providing a content-based signal for distinguishing meaningful merged regions from spatially larger regions with little or no content.

The objective is evaluated without access to ground-truth cell boundaries. This allows us to test whether a content-based selection criterion can recover candidates removed by the original merge rule.

Object Detection (TATR) TATR predicts table rows, columns, and cells as bounding boxes. Its post-processing stage uses confidence-based non-maximum suppression to remove overlapping detections.

We formulate the same selection problem as MWIS, with overlapping boxes represented as conflict edges. We evaluate two objectives:

1. **Model confidence:** $w_i = s_i$, where s_i is the model confidence score.
2. **Geometric regularity:** a score based on the deviation of the candidate height from the median height of the corresponding predictions.

The first objective tests whether global optimization can improve upon greedy confidence-based NMS. The second tests whether a geometric signal provides useful information when model confidence alone is insufficient.

Sequence Generation (TableFormer) TableFormer generates structural tokens and associated continuous cell coordinates. Its post-processing stage performs coordinate-based matching and deduplication.

For the optimization experiment, candidates are represented as nodes in a conflict graph. Because small coordinate differences can create artificial intersections between otherwise equivalent predictions, we use an IoU-based conflict criterion rather than treating any non-zero coordinate overlap as a conflict. Specifically, two candidates are connected when their IoU exceeds the selected conflict threshold.

The objective and threshold are evaluated without access to ground-truth annotations.

Relation Learning (LORE-TSR) LORE predicts spatial bounding boxes and logical row/column coordinates using an anchor-free CenterNet-based detection head. Source tracing shows that a substantial fraction of the observed cell losses occur when detection peaks fail to cross the confidence threshold.

These candidates therefore never enter a subsequent reconstruction stage. Since there is no post-processing candidate-selection operator corresponding to the interventions tested in the other paradigms, we do not apply H2 to LORE.

3.6 Statistical Analysis

For experiments where multiple tables are evaluated independently, we compare the per-table true-positive counts before and after optimization. Statistical tests are applied only to experiments for which the required paired observations are available.

For the TableFormer analysis, which uses $N = 100$ tables, we report:

- a two-sided paired t-test;
- a Wilcoxon signed-rank test; and
- a percentile bootstrap confidence interval based on 10,000 resamples.

The bootstrap interval is computed for the mean per-table difference

$$\Delta\mathrm{TP} = \mathrm{TP}_{\mathrm{opt}} - \mathrm{TP}_{\mathrm{raw}}. \tag{9}$$

We describe an outcome as statistically indistinguishable from the raw predictions when the corresponding hypothesis tests do not reject the null hypothesis at $p < 0.05$ and the bootstrap confidence interval includes zero.

These statistical tests are used to quantify the observed differences; they are not used to determine whether a particular optimization formulation is theoretically valid.

4 Results

We evaluate the four TSR paradigms using the framework described in Section 3. The experiments produce four distinct outcomes: substantial recovery for SPLERGE, limited benefit for TATR, a statistically null result for TableFormer after correcting a methodological artifact, and non-applicability for LORE-TSR. These results show that replacing a heuristic with an exact optimizer does not by itself guarantee better structural recovery.

Table 2: Results across four TSR paradigms. "Destroyed" denotes true-positive predictions present before post-processing but absent afterward. The optimization result is reported for the corresponding exact intervention described in Section 3.5.

Paradigm	Raw TP	Post-processed TP	Destroyed TP	Optimized TP	Outcome
SPLERGE	1,833	1,203	634	1,836	Positive
TATR	1,244	–	7 rows	1,245	Limited
TableFormer	5,204	–	–	5,203	Statistically null
LORE-TSR	–	–	68.5% of cell losses	N/A	Not applicable

4.1 Overall Results

Table 2 summarizes the main findings. The results are reported at the prediction level for each paradigm; absolute values should not be interpreted as directly comparable across datasets or prediction granularities.

The four experiments differ not only in the magnitude of the observed changes but also in the mechanism responsible for them. SPLERGE provides a discrete candidate space in which the reconstruction stage removes many correct candidates. TATR already has a conflict structure for which greedy confidence-based NMS is nearly equivalent to exact selection. TableFormer produces continuous coordinate predictions, making the remaining errors less amenable to discrete selection. LORE-TSR exhibits losses before a reconstruction stage is reached.

4.2 SPLERGE: Substantial Recovery of Destroyed Predictions

SPLERGE provides the clearest positive result for H1 and H2. The raw split grid contains 1,833 true-positive cells, corresponding to a recall of 32.47% under the evaluation protocol used in the experiment. Applying the original merge heuristic reduces the count to 1,203 true positives, or 21.31% recall. Thus, 634 true-positive predictions present in the raw grid are removed during reconstruction.

Replacing the merge rule with exact set packing recovers 1,836 true positives, corresponding to 32.52% recall. Of the 634 true positives destroyed by the heuristic, 632 are recovered, giving a recovery rate of 99.68%. Across the evaluated tables, 74 improve after optimization, 26 remain unchanged, and none degrade.

The recovery depends on the objective used by the optimizer. A content-tightness term based on OCR token coverage is required to prevent the optimizer from favoring unnecessarily large merged regions. In particular, the selected objective rewards merges that explain fragmented content rather than simply increasing the spatial area covered by a candidate.

These results provide evidence that the original reconstruction rule can remove correct predictions and that, for this representation, global candidate selection can recover most of those predictions.

Checkpoint qualification. The absolute SPLERGE performance should not be interpreted as a benchmark of the published SPLERGE architecture. A later

source and literature audit showed that the checkpoint used in this experiment does not match the published SPLERGE architecture and has substantially lower raw performance than reported reproductions. The preprocessing pipeline was independently checked and found to be correct. We therefore interpret this experiment as a controlled stress test of the reconstruction mechanism rather than as a new performance result for SPLERGE. The qualitative observation of destructive merging remains the focus of the experiment.

4.3 TATR: Limited Benefit from Exact Selection

The TATR experiments show that exact optimization provides little additional benefit when the optimization objective largely reproduces the score used by greedy NMS.

Using the model confidence as the optimization objective increases the true-positive count only from 1,244 to 1,245. This behavior is explained by the structure of the conflict graph: 1,270 of 1,272 connected components (99.8%) contain only a pair of mutually conflicting candidates. For such components, selecting the higher-confidence candidate is already optimal under the same objective. Exact MWIS therefore has almost no additional freedom over greedy NMS.

We next replace model confidence with a geometric regularity objective based on deviation from the median row height. This objective is independent of the detection confidence and therefore tests whether a different signal can improve candidate selection. Across the full corpus, it produces a net gain of only one true positive: three correct predictions are recovered in the most affected case, while two correct predictions are lost elsewhere.

Source-level inspection identifies the same mechanism in the newly destroyed cases. A false duplicate row can be marginally closer to the median row height than the genuine row. In one case, this geometric preference overrides a substantial confidence difference between the candidates. Thus, changing the objective changes the failure mode rather than eliminating it.

The TATR results therefore indicate that the limitation is not the ability of the optimizer to solve the selection problem. Rather, the available objective does not reliably distinguish correct from incorrect candidates.

4.4 LORE-TSR: Failure Before the Reconstruction Stage

LORE-TSR represents a different failure mode. Source tracing shows that 68.5% of observed cell losses occur when CenterNet activation peaks fail to cross the detection threshold. These cells are therefore never emitted as candidates for a subsequent reconstruction stage.

Because the relevant structural loss occurs before the candidate-selection stage, there is no post-processing operator that can be replaced by the set-packing formulation used in the other experiments. H2 is consequently not applicable to this failure mode.

Table 3: TableFormer statistical comparison. Differences are computed between optimized and raw predictions over 100 tables.

Statistic	Value
Raw true positives	5,204
Optimized true positives	5,203
Tables improved	0
Tables tied	99
Tables degraded	1
Paired t-test	$p = 0.320$
Wilcoxon signed-rank test	$p = 0.317$
Bootstrap 95% CI	$[-0.030, 0.000]$

This result also provides an important boundary for the proposed analysis: post-processing optimization can only select among candidates that the underlying model has already produced.

4.5 TableFormer: Statistically Null After Correcting a Conflict Artifact

The initial TableFormer optimization experiment produced an apparent loss of 190 true positives relative to the raw predictions, decreasing the count from 5,204 to 5,014. Source tracing showed that this result was caused by the conflict definition rather than by the optimization formulation itself. Because TableFormer predicts continuous coordinates, two otherwise valid boxes can exhibit a small non-zero geometric intersection due to sub-pixel regression error. One such case involved only 0.24 pixels of edge overlap.

We therefore replaced the strict non-zero-intersection rule with an IoU-based conflict criterion and evaluated 28 tolerance settings. The apparent 190-cell gap disappeared under tolerant conflict definitions, with 98.9–99.5% of the apparent loss recovered. At an IoU threshold of 0.5, the optimized prediction contains 5,203 true positives compared with 5,204 in the raw predictions.

The remaining one-cell difference is not statistically significant. Across 100 tables, 99 tables are tied, one table loses a true positive, and no table improves. The paired t-test gives $p = 0.320$, while the Wilcoxon signed-rank test gives $p = 0.317$. The 10,000-sample bootstrap confidence interval for the mean difference is $[-0.030, 0.000]$.

The result is therefore statistically indistinguishable from the raw predictions. The optimization does not produce a measurable net gain, but it also does not reproduce the catastrophic loss observed under the original conflict definition. The remaining errors are primarily associated with continuous coordinate drift rather than a discrete ambiguity that can be resolved by selecting one candidate over another.

4.6 Cross-Paradigm Analysis

The four experiments indicate that optimization benefit depends on both the candidate representation and the selection objective. SPLERGE provides a discrete candidate space together with an OCR-based content signal, allowing global selection to recover many predictions removed by the merge heuristic. In TATR, the confidence objective largely reproduces greedy NMS, while the alternative geometric objective does not reliably distinguish correct from incorrect candidates. TableFormer's continuous coordinates limit what selection can recover, and LORE illustrates the complementary case in which the relevant prediction is never emitted. Thus, exact optimization can improve post-processing only when the raw candidate set and its scoring signal contain sufficient information for a better structural selection.

5 Conclusion

We presented a cross-paradigm empirical study of post-processing in Table Structure Recognition (TSR). By separating raw model predictions from subsequent reconstruction and filtering, we examined whether correct structural predictions can be lost after model inference and whether exact combinatorial optimization can recover them. Across four TSR paradigms, the experiments produced four distinct outcomes rather than a uniform benefit from replacing heuristics.

The SPLERGE experiment showed the clearest recovery: exact set packing recovered 632 of 634 true positives removed by the original merge procedure. In TATR, exact MWIS provided little additional benefit because most conflict components were simple pairwise conflicts and the confidence objective reproduced the behavior of greedy NMS. An alternative geometric objective produced only a net gain of one true positive. For TableFormer, correcting a continuous-coordinate conflict artifact reduced an apparent large performance loss to a statistically indistinguishable result from the raw predictions. LORE-TSR exposed a different limitation, with many failures occurring at the neural detection stage before a post-processing candidate set was formed.

Taken together, these results suggest that the usefulness of exact optimization depends on both the representation of the candidate space and the information available to score competing candidates. A discrete candidate space provides explicit alternatives for global selection, while a content-aware objective can provide information beyond the local confidence or geometric rules used by existing heuristics. When either component is absent, exact optimization provides limited or no additional recovery. More broadly, the study shows that evaluating only the final TSR output can obscure errors introduced between model prediction and structural reconstruction. Explicitly tracing this transition provides a way to distinguish errors made by the model from errors introduced by post-processing.

References

1. Chi, Z., Huang, H., Xu, H.D., Yu, H., Yin, W., Mao, X.L.: Complicated table structure recognition. arXiv preprint arXiv:1908.04729 (2019) 2, 5
2. Guo, Z., Yu, Y., Lv, P., Zhang, C., Li, H., Wang, Z., Yao, K., Liu, J., Wang, J.: TRUST: An accurate and end-to-end table structure recognizer using splitting-based transformers. arXiv preprint arXiv:2208.14687 (2022) 4
3. Huang, Y., Lu, N., Chen, D., Li, Y., Xie, Z., Zhu, S., Gao, L., Peng, W.: Improving table structure recognition with visual-alignment sequential coordinate modeling. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 11134–11143 (2023) 4
4. Lin, W., Sun, Z., Ma, C., Li, M., Wang, J., Sun, L., Huo, Q.: TSRFormer: Table structure recognition with transformers. In: Proceedings of the 30th ACM International Conference on Multimedia. ACM (2022). https://doi.org/10.1145/3503161.3548391 4
5. Lyu, P., Ma, W., Wang, H., Yu, Y., Zhang, C., Yao, K., Xue, Y., Wang, J.: GridFormer: Towards accurate table structure recognition via grid prediction. In: Proceedings of the 31st ACM International Conference on Multimedia. pp. 7747–7757. ACM (2023). https://doi.org/10.1145/3581783.3611961 4
6. Nassar, A., Livathinos, N., Lysak, M., Staar, P.: Tableformer: Table structure understanding with transformers. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 4614–4623 (2022) 2, 4
7. Qiao, L., Li, Z., Cheng, Z., Zhang, P., Pu, S., Niu, Y., Ren, W., Tan, W., Wu, F.: LGPMA: Complicated table structure recognition with local and global pyramid mask alignment. In: Document Analysis and Recognition – ICDAR 2021. pp. 99–114. Springer (2021). https://doi.org/10.1007/978-3-030-86549-8_7 4
8. Raja, S., Mondal, A., Jawahar, C.V.: Table structure recognition using top-down and bottom-up cues. In: Computer Vision – ECCV 2020. pp. 70–86. Springer (2020). https://doi.org/10.1007/978-3-030-58604-1_5 4
9. Schreiber, S., Agne, S., Wolf, I., Dengel, A., Ahmed, S.: Deepdesrt: Deep learning for detection and structure recognition of tables in document images. In: 2017 14th IAPR International Conference on Document Analysis and Recognition (ICDAR). pp. 1162–1167. IEEE (2017). https://doi.org/10.1109/ICDAR.2017.192 4
10. Smock, B., Pesala, R., Abraham, R.: Pubtables-1m: Towards comprehensive table extraction from unstructured documents. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 4634–4642 (2022) 1, 2, 4, 5
11. Tensmeyer, C., Morariu, V.I., Price, B., Cohen, S., Martinez, T.: Deep splitting and merging for table structure decomposition. In: 2019 International Conference on Document Analysis and Recognition (ICDAR). pp. 114–121. IEEE (2019). https://doi.org/10.1109/ICDAR.2019.00027 4
12. Xiao, A., Yang, C.: Towards one-stage end-to-end table structure recognition with parallel regression for diverse scenarios. arXiv preprint arXiv:2504.17522 (2025) 5
13. Xing, H., Gao, F., Long, R., Bu, J., Zheng, Q., Li, L., Yao, C., Yu, Z.: LORE: Logical location regression network for table structure recognition. In: Proceedings of the AAAI Conference on Artificial Intelligence. vol. 37, pp. 2992–3000 (2023). https://doi.org/10.1609/AAAI.V37I3.25402 2, 5
14. Xue, W., Yu, B., Wang, W., Tao, D., Li, Q.: TGRNet: A table graph reconstruction network for table structure recognition. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV). pp. 1295–1304 (2021) 2, 5

15. Zhang, Z., Zhang, J., Du, J., Wang, F.: Split, embed and merge: An accurate table structure recognizer. Pattern Recognition **126**, 108565 (2022). https://doi.org/10.1016/j.patcog.2022.108565 2, 4
16. Zheng, X., Burdick, D., Popa, L., Zhong, X., Wang, N.X.R.: Global table extractor (gte): A framework for joint table identification and cell structure recognition using visual context. In: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV). pp. 697–706 (2021) 4
17. Zhong, X., ShafieiBavani, E., Jimeno Yepes, A.: Image-based table recognition: Data, model, and evaluation. In: Computer Vision – ECCV 2020. pp. 564–580. Springer (2020). https://doi.org/10.1007/978-3-030-58598-3_34 1, 4, 5